



Problem Challenge 1

We'll cover the following ^

- Reconstructing a Sequence (hard)
- Try it yourself

Reconstructing a Sequence (hard)

Given a sequence `originalSeq` and an array of sequences, write a method to find if `originalSeq` can be uniquely reconstructed from the array of sequences.

Unique reconstruction means that we need to find if `originalSeq` is the only sequence such that all sequences in the array are subsequences of it.

Example 1:

Input: `originalSeq: [1, 2, 3, 4]`, `seqs: [[1, 2], [2, 3], [3, 4]]`

Output: `true`

Explanation: The sequences `[1, 2]`, `[2, 3]`, and `[3, 4]` can uniquely reconstruct `[1, 2, 3, 4]`, in other words, all the given sequences uniquely define the order of numbers in the '`originalSeq`'.

Example 2:

Input: `originalSeq: [1, 2, 3, 4]`, `seqs: [[1, 2], [2, 3], [2, 4]]`

Output: `false`

Explanation: The sequences `[1, 2]`, `[2, 3]`, and `[2, 4]` cannot uniquely reconstruct `[1, 2, 3, 4]`. There are two possible sequences we can construct from the given sequences:

- 1) `[1, 2, 3, 4]`
- 2) `[1, 2, 4, 3]`

Example 3:

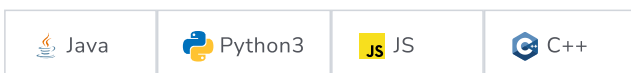
Input: `originalSeq: [3, 1, 4, 2, 5]`, `seqs: [[3, 1, 5], [1, 4, 2, 5]]`

Output: `true`

Explanation: The sequences `[3, 1, 5]` and `[1, 4, 2, 5]` can uniquely reconstruct `[3, 1, 4, 2, 5]`.

Try it yourself

Try solving this question here:



```
1 import java.util.*;
2
3 class SequenceReconstruction {
4     public static boolean canConstruct(int[] originalSeq, int[][] sequences) {
5         // TODO: Write your code here
6         return false;
7     }
8
9     public static void main(String[] args) {
10         boolean result = SequenceReconstruction.canConstruct(new int[] { 1, 2, 3, 4 },
11             new int[][] { new int[] { 1, 2 }, new int[] { 2, 3 }, new int[] { 3, 4 } });
12         System.out.println("Can we uniquely construct the sequence: " + result);
13
14         result = SequenceReconstruction.canConstruct(new int[] { 1, 2, 3, 4 },
15             new int[][] { new int[] { 1, 2 }, new int[] { 2, 3 }, new int[] { 2, 4 } });
16         System.out.println("Can we uniquely construct the sequence: " + result);
17
18         result = SequenceReconstruction.canConstruct(new int[] { 3, 1, 4, 2, 5 },
19             new int[][] { new int[] { 3, 1, 5 }, new int[] { 1, 4, 2, 5 } });
20         System.out.println("Can we uniquely construct the sequence: " + result);
21     }
22 }
```

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Alien Dictionary (hard)

Solution Review: Problem Challenge 1